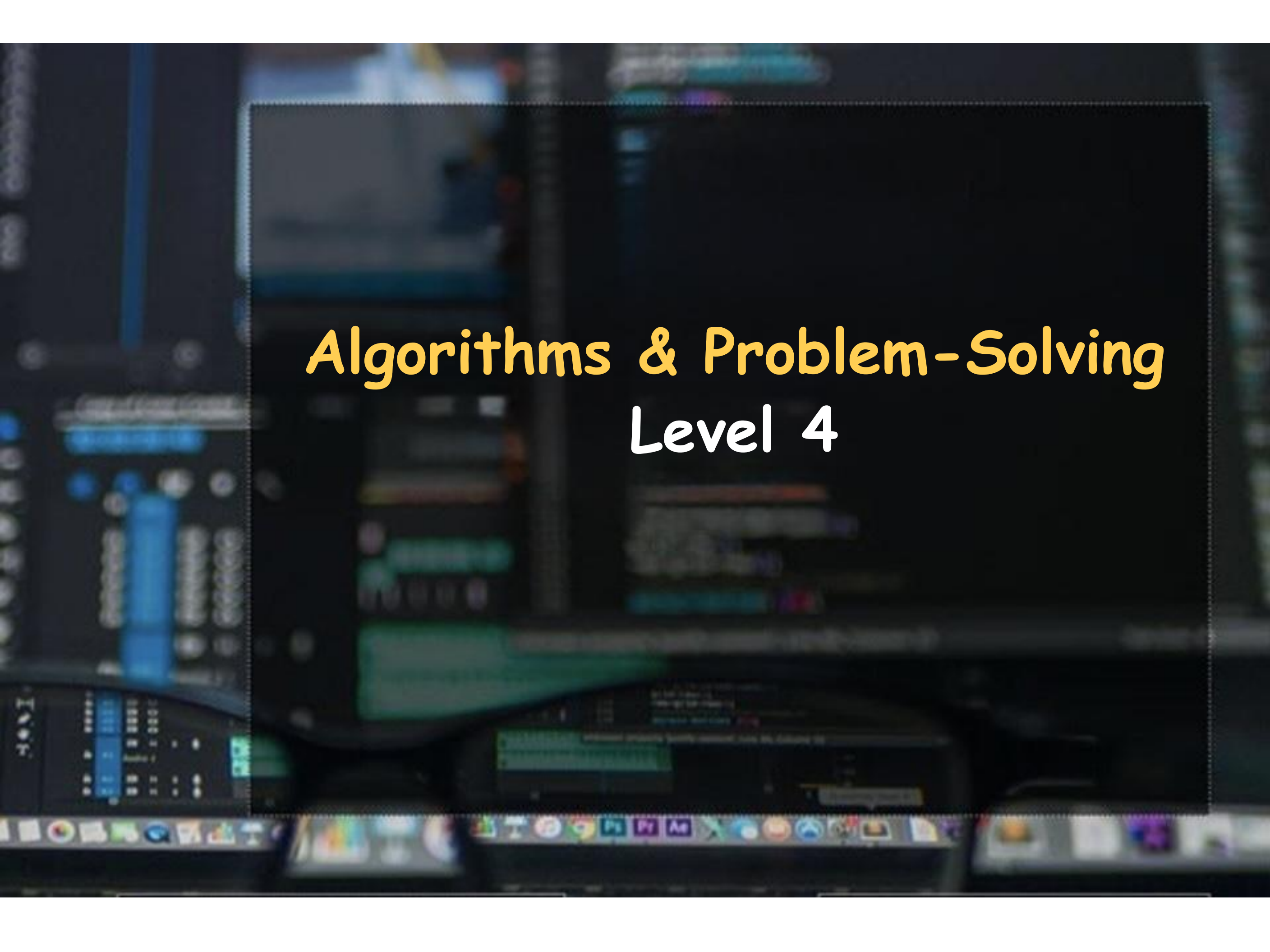
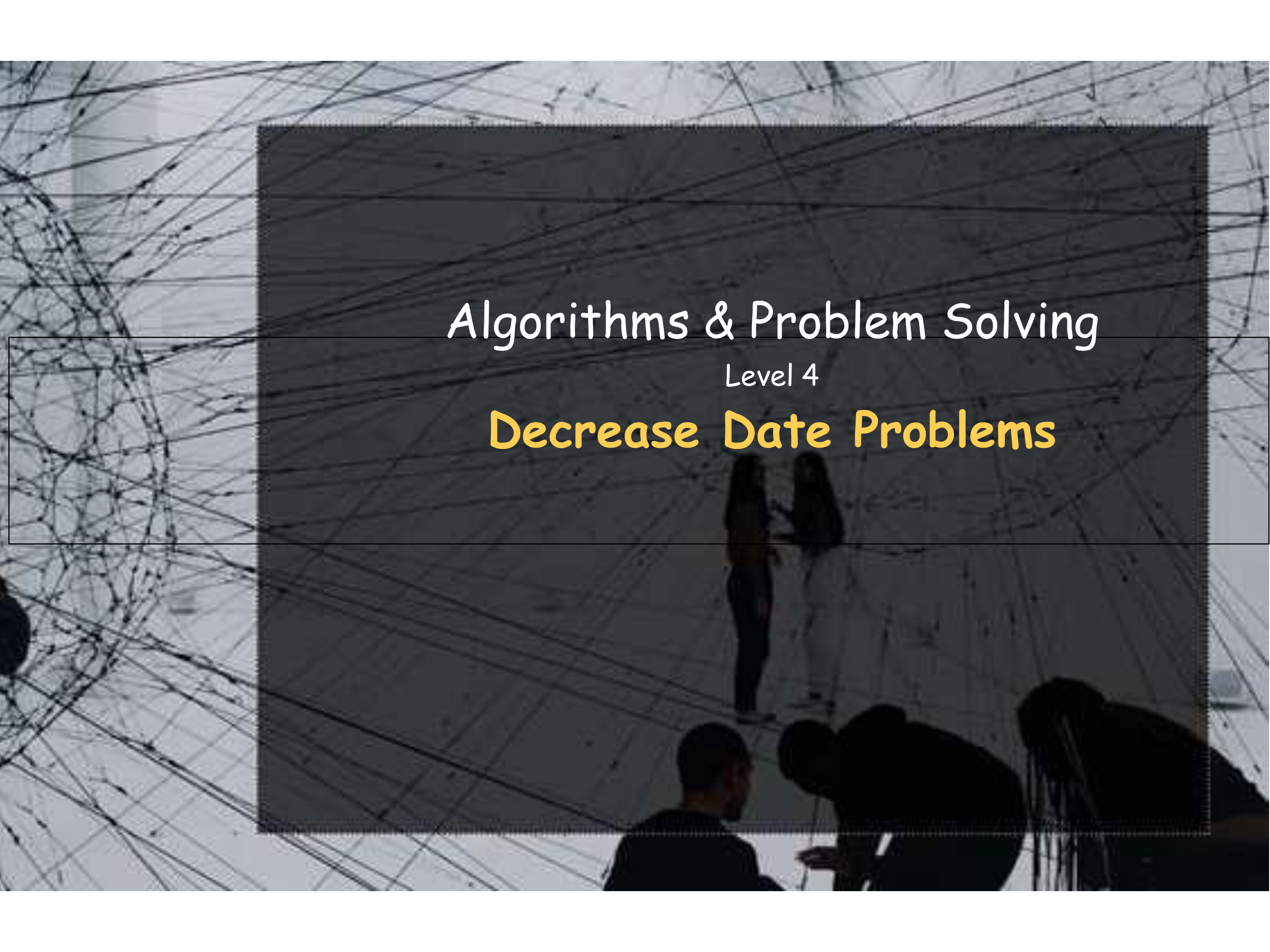




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, which adds a human element to the technical theme.

Algorithms & Problem Solving

Level 4

Decrease Date Problems



Definition

- The program's objective is to read a date from the user and then subtract from it a specified amount of days, weeks, months, years, decades, centuries, or millennia, while maintaining the correct logic for transitioning between months and years.

Procedures

- 1 Defining the sDate structure:
 - It contains three elements: `Day`, `Month`, and `Year` to represent the date.
- 2 Reading input from the user:
 - ReadDay() - Reads the day
 - ReadMonth() - Reads the month
 - ReadYear() - Reads the year
 - All are combined into the ReadFullDate() function to return the full date.
- 3 Checking for leap years:
 - The IsLeapYear(year) function checks if the year is a leap year (i.e., February has 29 days).
- 4 Number of days in each month:
 - NumberOfDaysInAMonth(month, year) returns the correct number of days in the month, taking into account leap years.
- 5 Decrease by one day:
 - DecreaseDateByOneDay(date) decreases the date by one and automatically adjusts. Return to the previous month or year when you reach the beginning of the month or year.
 - Example: From `1/1/2023` → `31/12/2022`

- 6 Decrease a specific number of days or weeks:
 - DecreaseDateByXDays(days, date) i Repeat decrementing by one day the number of days required
 - DecreaseDateByOneWeek(date) i Decrease by 7 days
 - DecreaseDateByXWeeks(weeks, date) i Repeat the process the number of weeks required
- 7 Decrease a month or several months:
 - DecreaseDateByOneMonth(date) i Decrease by one month, adjusting the day if it doesn't exist (e.g., going from March 31st to February 28th)
 - DecreaseDateByXMonths(months, date) i Repeat the process for several months
- 8 Decrease a year or more:
 - DecreaseDateByOneYear(date) i Decrease by one year and treat February 29th in non-years Leap Days
 - DecreaseDateByXYears(years, date)` repeats the process year after year.
 - DecreaseDateByXYearsFaster(years, date)` reduces the number of years directly in one step.
- 9 Decades, Centuries, and Millennia:
 - DecreaseDateByOneDecade()` reduces the number of years by 10.
 - DecreaseDateByXDecades()` reduces the number of years by 10 (decade x 10) repeatedly.
 - DecreaseDateByXDecadesFaster()` reduces the number of years by 10.
 - DecreaseDateByOneCentury()` reduces the number of years by 100.
 - DecreaseDateByOneMillennium()` reduces the number of years by 1000.



Example:

Please enter a Day? 31

Please enter a Month? 12

Please enter a Year? 2022

Date After:

01: Subtracting one day	is: 30/12/2022
02: Subtracting 10 days	is: 20/12/2022
03: Subtracting one week	is: 13/12/2022
04: Subtracting 10 weeks	is: 04/10/2022
05: Subtracting one month	is: 04/09/2022
06: Subtracting 5 months	is: 04/04/2022
07: Subtracting one year	is: 04/04/2021
08: Subtracting 10 years	is: 04/04/2011
09: Subtracting 10 years (faster)	is: 04/04/2001
10: Subtracting one Decade	is: 04/04/1991
11: Subtracting 10 Decades	is: 04/04/1891
12: Subtracting 10 Decades (faster)	is: 04/04/1791
13: Subtracting one Century	is: 04/04/1691
14: Subtracting one Millennium	is: 04/04/0691



Note:

- The "Faster" functions perform calculations directly on the year, saving time and eliminating repetition.
- The program maintains the accuracy of the day even when switching to a month with fewer days.



Comparison:

Type

Slow (loop)

Fast (direct)

Method

Repeats the process multiple times

Reduces years directly

Performance

Accurate but slower

Faster but requires
additional verification from
February 29th

Summary

- ✓ The program demonstrates a comprehensive application of date handling concepts.
- ✓ It covers all possible time units for deduction.
- ✓ It maintains accurate date logic when switching between months and years.

Thank you 😊